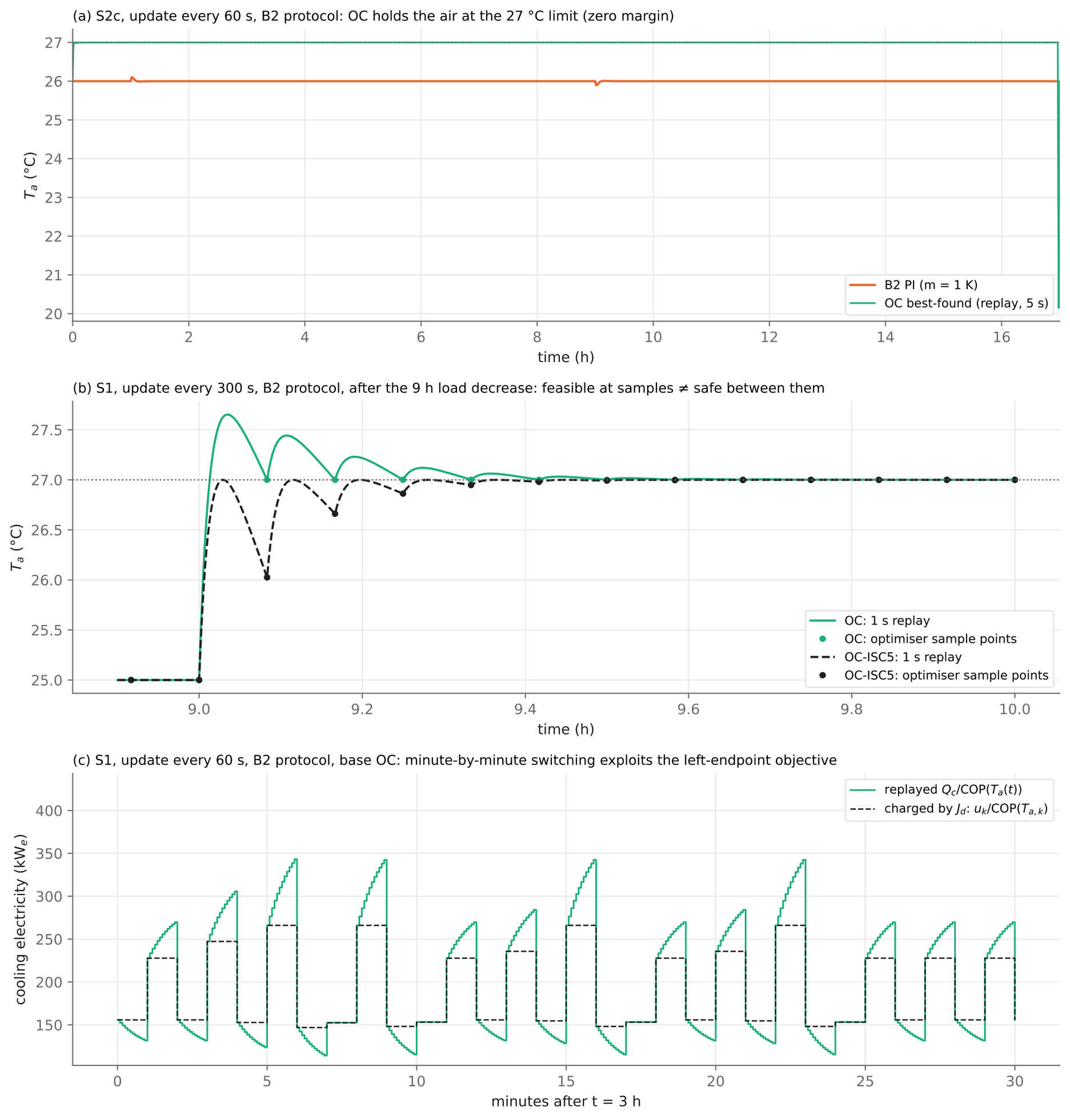
(a) S2c, update every 60 s, B2 protocol: OC holds the air at the 27 °C limit (zero margin)
T_a (°C)
time (h)
B2 PI (m = 1 K)
OC best-found (replay, 5 s)
(b) S1, update every 300 s, B2 protocol, after the 9 h load decrease: feasible at samples ≠ safe between them
T_a (°C)
time (h)
OC: 1 s replay
OC: optimiser sample points
OC-ISC5: 1 s replay
OC-ISC5: optimiser sample points
(c) S1, update every 60 s, B2 protocol, base OC: minute-by-minute switching exploits the left-endpoint objective
cooling electricity (kW$_e$)
minutes after t = 3 h
replayed Q_c/COP($T_a(t)$)
charged by J_d: u_k/COP($T_{a,k}$)